

Overview of Agentic Workflow

1.

User prompts coding agent

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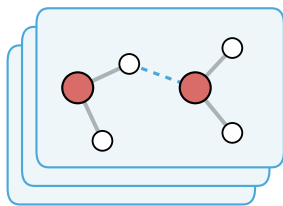
Run full-agentic-workflow on the 350 dimer geometries in @dimer_dataset/. Use the
> CCSD(T)/CBS references from @SI.pdf and a 2.5 hour walltime budget. Consider all levels of theory in the SI, and show me the accuracy options before submitting calculations.

dir ./dimer_dataset/

pdf ./SI.pdf

2.

Agent calls error & time estimator models



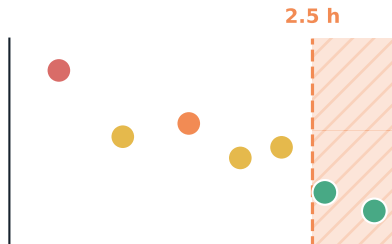
Geometries



Methods, bases, BSSE correction



pred. % Error



pred. Wall Time

< 2%	HIGH
2-5%	MED
5-10%	LOW
≥ 10%	—

3.

User selects computations

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◆ Predictions are ready for your review, which calculations should we submit?

☒ PBE-D3 / aTZ 1.4% 38 min

☒ wB97X-V / aTZ 1.8% 1.6 h

☐ B3LYP-D3 / aTZ 3.2% 52 min

☐ MP2 / aTZ 7.1% 2.1 h

☐ HF / aTZ 18% 12 min

☐ B2PLYP-D3 / aTZ 4.4% 2.6 h

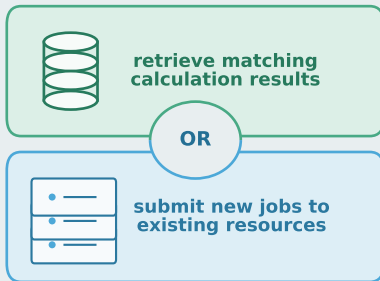
> Let's run PBE-D3, B3LYP-D3, and MP2 on all systems.

4.

Agent connects to server



Agent connects to QCfractal server



Checkpoint files created

5.

User prompts agent to query results.

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◆ All jobs submitted and running, checkpoint files saved.

+ checkpoint.csv
+ recommendations.md

>

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> Check job status.

25 jobs complete, 13 errored, and 1012 still running.

> Re-queue errored jobs.

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> Status?

◆ All jobs complete.

> Provide a detailed summary of results.